

DLD lab quiz

01.

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

A	B	G_{n+1}
0	0	1
0	1	G'
1	0	G
1	1	0

$$\therefore B = K, \quad J = A'$$

Ans: 1

02.

$$f(A, B, C, D) = \sum(0, 1, 4, 6, 7, 8, 10, 12, 14, 15)$$

A	B	C	D	f
0	0	0	0	1
0	0	0	1	1 ✓
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1 ✓
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

A	B	C	D	
0	0	0	1	→ 1 ✓
0	1	0	1	→ 9 Set
1	0	0	1	→ 9 x
1	1	0	1	→ 13 x

$$\overline{A}B = \sim(A+B)$$

$$\overline{A}B \neq \overline{A+B}$$

Conferent from not gives

03.

$$\text{No } \lceil \log_2 N \rceil = \lceil \log_2 5 \rceil$$

$$= 3$$

(Ans)

04.

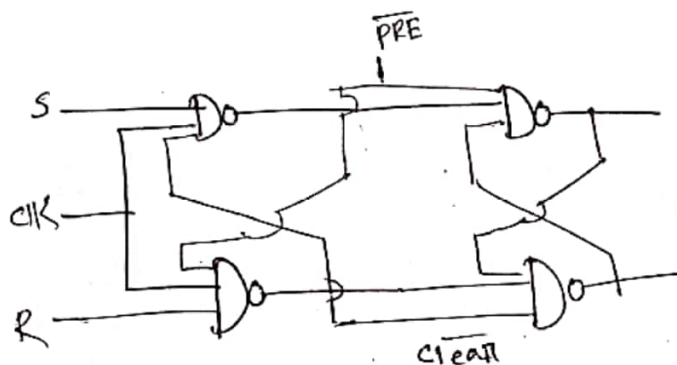
$$y' = (a(bcd)'e)'$$

$$y = a(bcd)'e$$

Q $y=1$, if, $a=e=1$, $bcd \neq 1$

Ans. $a=e=1$, $b=c=d=0$

05.



Q Ans: 4

06.

No. of enablers in 74138 $\rightarrow 3$

07.

Equivalence gate — Ex-NOR $\oplus$

(if bit same, output 1)
 $0 \oplus 0 = 1$, $1 \oplus 1 = 1$

08.

(M) In two adjacent codes, there is change in one bit.

09.

(i)

✓ (ii) True

(iii)

✓ (iv) D most economical (in terms of transistor count and silicon area)

10.



$$Y = A_2 \bar{A}_1 + \bar{A}_2 A_1$$

11.

4 to 1 mux

s_1	s_0	Y_0
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$$Y_0 = \bar{s}_0 \bar{s}_1 I_0 + \bar{s}_1 s_0 I_1 + s_1 \bar{s}_0 I_2 + s_1 s_0 I_3$$

$2 + 1 + 2 + 1 + 2 + 1 + 2 + 1 = 13$

12.

$$\frac{11}{3} \quad \frac{1001}{9}$$

(Ans) 39

13. $f(x, y, z) = (x \rightarrow y) \rightarrow z$

14. Out of syllabus

15. 7483 is a high speed 4-bit full adder IC.

16. 1, Active low

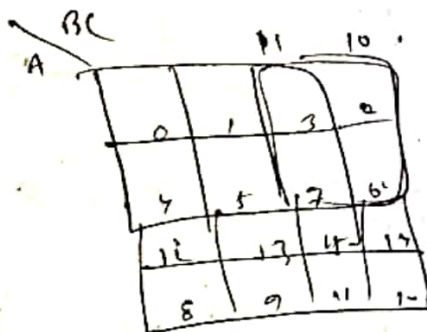
17.

18. 0, 1, 0, 2,

Ans: 2

00
01
00
10

19. $F = \sum (2, 3, 6, 7)$
 $= B$



20.

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$\overline{\overline{A}B + A\overline{B}} = \overline{\overline{A}B} \cdot \overline{A\overline{B}} = \overline{\overline{A}} \cdot \overline{B} = AB$$

$$= \overline{A+B} + \overline{A+B}$$

$$= \overline{A+B} + \overline{A+B}$$

$$= (\overline{A+B}) \cdot (\overline{A+B}) = \overline{(A+B) + (A+B)}$$

$$A \oplus B = \overline{\overline{A}B + A\overline{B}}$$

$$= \overline{\overline{A}B} \cdot \overline{A\overline{B}}$$

$$= \overline{\overline{A}B} \cdot \overline{A\overline{B}}$$

Am 6

21.

		CD			
AB		00	01	11	10
	00	1			1
	01	1			
	11				1
	10	1			1

$$F = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}D$$

$$= \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}D$$

21.

if decimal 12 $\rightarrow$ $\begin{array}{r} \text{exam 3} \\ 1 \quad 2 \\ +3 \quad +3 \\ \hline 0100 \quad 0101 \end{array}$

if dec 12 $\rightarrow$ don't care

22.

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

23. ~~Minterms~~ K-map

24. $P(S(T+R))$

25. 0,0,1,1

00
00
01
01

$$\lceil \log_2 4 \rceil = 2$$

26.

$$2^3 = 8$$

25.

0, 2, 4, 8, 12, 14

AB \ CD	00	01	11	10
00	-	1	1	-
01	-	1	1	1
11	1	1	1	-
10	-	1	1	-

$$F' = D + \bar{A}BC + AB\bar{C}$$

$$F = D' (A+B+C) (A'+B'+C) \quad (A)$$

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	0	0
11	0	0	1	0
10	1	0	1	0

28.

AB \ CD	00	01	11	10
00	-	-	-	-
01	-	-	-	-
11	-	-	-	-
10	-	-	-	-

AB \ CD	00	01	11	10
00	-	-	-	-
01	-	-	-	-
11	-	-	-	-
10	-	-	-	-

2 variables

29.

0-7

$$\lceil \log_2 N \rceil = 3$$

30.

Diagonal

$$8 = 2^3$$

31.

$$AB + A'C + BC$$

$$= AB + C(A' + B)$$

$$ABC' + ABC + A'BC + A'B'C + \underline{ABC} + \underline{A'BC}$$

$$= \underline{ABC} + \underline{ABC'} + \underline{A'BC'} + \underline{A'B'C} + \underline{A'BC}$$

$$= AB + \underline{A'B} + \underline{A'B'C} = ABC + A'BC + ABC' + A'B'C$$

$$= B(A + A') + A'B'C = AB(C + C') + A'C(B' + B)$$

$$= \underline{B} + \underline{A'B'C} = AB + A'C$$

32. active low,

33,

34. ~~1000~~ 1001 $\rightarrow$ 1000

$\begin{cases} 0100 \rightarrow \text{Rishu} \\ 1000 \rightarrow \text{Wib} \end{cases}$

Q 2

31.

(P)	Q	(R)
0	0	0
0	1	1
1	0	1
1	1	0

32. $P \oplus Q = R$

$P \oplus R = Q$

33. True